WS N°1: Diffusion

1) A sheet of steel 1.5 mm thick has nitrogen atmospheres on both sides at $1200°C$ and is permitted to achieve a steady-state diffusion condition. The diffusion coefficient for nitrogen in steel at this temperature is $6 \times 10^{-11}\, m^2/s$, and the diffusion flux is found to be $1.2 \times 10^{-7}\, kg/m^2.s$. Also, it is known that the concentration of nitrogen in the steel at the high-pressure surface is $4\, kg/m^3$. How far into the sheet from this high-pressure side will the concentration be is $2\, kg/m^3$? Assume a linear concentration profile.

Solution:

$$J = -D\frac{C_A - C_B}{x_A - x_B}$$

$C_A = 4\, kg/m^3$
Assume x_A to be at the surface
⇨ $x_A = 0$

$C_B = 2\, kg/m^3, x_B = ?$

$D = 6 \times 10^{-11}\, m^2/s$
$J = 1.2 \times 10^{-7}\, kg/m^2.s$

⇨ $x_B = x_A + D\frac{C_A - C_B}{J} = 0 + \frac{(6\times10^{-11} m^2/s)\times(4-2)4kg/m^3}{1.2\times10^{-7} kg/m^2.s} = 1 \times 10^{-3} m = 1mm$

2) The purification of hydrogen gas by diffusion through a palladium sheet is performed. Compute the number of kilograms of hydrogen that pass per hour through a $5mm$ thick sheet of palladium having an area of $0.2\, m^2$ at $500°C$. Assume a diffusion coefficient of $1 \times 10^{-8}\, m^2/s$, that the concentrations at the high- and low-pressure sides of the plate are 2.4 and 0.6 kg of hydrogen per cubic meter of palladium, and that steadystate conditions have been attained.

Solution:

$$J = \frac{M}{At}$$

⇨ $\frac{M}{t} = J \times A = -D\frac{\Delta C}{\Delta x} \times A = -1 \times 10^{-8}\, m^2/s \times \frac{(2.4-0.6)kg/m^3}{-5\times10^{-3} m} \times 0.2\, m^2 =$
$7.2 \times 10^{-7}\, kg/s \times 3600\, s/hr$

⇨ $\frac{M}{t} = 2.59 \times 10^{-3}\, kg/hr$

3) A sheet of BCC iron 1 mm thick was exposed to a carburizing gas atmosphere on one side and a decarburizing atmosphere on the other side at $725°C$. After having reached steady state, the iron was quickly cooled to room temperature. The carbon

concentrations at the two surfaces of the sheet were determined to be 0.012 and 0.0075 $wt\%$. Compute the diffusion coefficient if the diffusion flux is $1.4 \times 10^{-8}\ kg/m^2.s$. The densities of carbon and iron are respectively 2.25 g/cm^3 and 7.87 g/cm^3.

Solution:
First, we have to convert the concentrations from $wt\%$ to kg/m^3.

For 0.012 $wt\%$: $C_C = 0.012wt\%$, $C_{Fe} = 100 - C_C = 100 - 0.012 = 99.988wt\%$

$$C_A = \frac{C_C}{\frac{C_C}{\rho_C} + \frac{C_{Fe}}{\rho_{Fe}}} = \frac{0.012wt\%}{\frac{0.012wt\%}{2.25\ g/cm^3} + \frac{99.988wt\%}{7.87\ g/cm^3}} \times \frac{10^3 kg/m^3}{1\ g/cm^3} = 0.941\ kg/m^3$$

For 0.0075 $wt\%$: $C_C = 0.0075wt\%$, $C_{Fe} = 100 - C_C = 100 - 0.0075 = 99.9925wt\%$

$$C_B = \frac{C_C}{\frac{C_C}{\rho_C} + \frac{C_{Fe}}{\rho_{Fe}}} = \frac{0.0075wt\%}{\frac{0.0075wt\%}{2.25\ g/m^3} + \frac{99.9925wt\%}{7.87\ g/m^3}} \times 10^3 kg/g = 0.590\ kg/m^3$$

$$J = -D\frac{C_A - C_B}{x_A - x_B}$$

$$x_A - x_B = -1mm = -1 \times 10^{-3}m$$

$$J = 1.4 \times 10^{-8}\ kg/m^2.s$$

$$D = -J\frac{x_A - x_B}{C_A - C_B} = \frac{-(1.4 \times 10^{-8}\ kg/m^2.s) \times (-1 \times 10^{-3}m)}{(0.941 - 0.590)\ kg/m^3} = 3.99 \times 10^{-11} m^2/s$$

4) Determine the carburizing time necessary to achieve a carbon concentration of 0.45 $wt\%$ at a position 2 mm into an iron–carbon alloy that initially contains 0.2 $wt\%$ C. The surface concentration is to be maintained at 1.3 $wt\%$ C, and the treatment is to be conducted at $1000°C$. Use the diffusion data for $\gamma - Fe$ from the table.

Solution:

$C_x = 0.45\ wt\%$ at $x = 2mm = 2 \times 10^{-3}m$

$$C_0 = 0.2\ wt\%$$

$$C_s = 1.3\ wt\%$$

$$\frac{C_x - C_0}{C_s - C_0} = 1 - \text{erf}\left(\frac{x}{2\sqrt{Dt}}\right)$$

$D = D_0 \exp\left(-\frac{Q_d}{RT}\right)$

Using the table of diffusion of C in $\gamma - Fe$:

$D_0 = 2.3 \times 10^{-5}\, m^2/s$ and $Q_d = 148KJ/mol = 148000\, J/mol$

$T = 1000°C + 273 = 1273K$

$$D = (2.3 \times 10^{-5}\, m^2/s\,) \times \exp\left(-\frac{148000\frac{J}{mol}}{8.31\, J/mol.k \times 1273K}\right) = 1.93 \times 10^{-11}\, m^2/s$$

$$\frac{0.45-0.2}{1.3-0.2} = 1 - \operatorname{erf}\left(\frac{2\times10^{-3}m}{2\sqrt{(1.93\times10^{-11}m^2/s)t}}\right) \Rightarrow \operatorname{erf}\left(\frac{2\times10^{-3}m}{2\sqrt{(1.93\times10^{-11}m^2/s)t}}\right) = 0.773$$

Using and interpolating from the table of error function values:

$$\frac{z-0.85}{0.9-0.85} = \frac{0.773-0.7707}{0.7970-0.7707} \Rightarrow z = 0.854 = \frac{2\times10^{-3}m}{2\sqrt{(1.93\times10^{-11}m^2/s)t}}$$

$\Rightarrow t = 71044 \sec = 19.73\, hrs$

5) For a steel alloy it has been determined that a carburizing heat treatment of 10 hrs duration will raise the carbon concentration to 0.45 $wt\%$ at a point 2.5 mm from the surface. Estimate the time necessary to achieve the same concentration at a 5 mm position for an identical steel and at the same carburizing temperature.

Solution:

Using Fick's 2nd law and to achieve the same concentration:

$$\frac{x^2}{Dt} = cte \Rightarrow \frac{{x_1}^2}{D_1 t_1} = \frac{{x_2}^2}{D_2 t_2}$$

Same carburizing temperature:

$D_1 = D_2$

- ⇨ $\frac{{x_1}^2}{t_1} = \frac{{x_2}^2}{t_2}$
- ⇨ $\frac{2.5^2}{10} = \frac{5^2}{t_2}$
- ⇨ $t_2 = 40\, hrs$

6) Calculate the values of the diffusion coefficients for the interdiffusion of carbon in both α-iron (BCC) and γ-iron (FCC) at $900°C$. Which is greater? Justify your answer.

Solution:

$$D = D_0 \exp\left(-\frac{Q_d}{RT}\right)$$

$$T = 900°C + 273 = 1173K$$

$$R = 8.31\,J/mol.k$$

Diffusion of C in $\alpha - Fe$ (BCC):
Using the table: $D_0 = 6.2 \times 10^{-7}\,m^2/s$ and $Q_d = 80KJ/mol = 80000\,J/mol$

$$D_\alpha = (6.2 \times 10^{-7}\,m^2/s\,) \times \exp\left(-\frac{80000\,\frac{J}{mol}}{8.31\,J/mol.k \times 1173K}\right) = 1.69 \times 10^{-10}\,m^2/s$$

Diffusion of C in $\gamma - Fe$ (FCC):
Using the table: $D_0 = 2.3 \times 10^{-5}\,m^2/s$ and $Q_d = 148KJ/mol = 148000\,J/mol$

$$D_\gamma = (2.3 \times 10^{-5}\,m^2/s\,) \times \exp\left(-\frac{148000\,\frac{J}{mol}}{8.31\,J/mol.k \times 1173K}\right) = 5.86 \times 10^{-12}\,m^2/s$$

⇨ $D_\alpha > D_\gamma$ ($D_{BCC} > D_{FCC}$)

⇨ The diffusion coefficient of C in $\alpha - Fe$ is greater than that in $\gamma - Fe$. This can be explained by the fact that the atomic packing factor in BCC is smaller than that in FCC ($APF_{BCC} < APF_{FCC}$)*. Therefore, there is slightly more interstitial void space in the BCC Fe, and, therefore, the motion of the interstitial carbon atoms occurs more easily (so greater diffusion coefficient in BCC).

7) At which temperature will the diffusion coefficient for the diffusion of copper in nickel have a value of $6.5 \times 10^{-17}\,m^2/s$.

Solution:

$$D = D_0 \exp\left(-\frac{Q_d}{RT}\right)$$

$$R = 8.31\,J/mol.k$$

Using the table of diffusion of copper in nickel:
$D_0 = 2.7 \times 10^{-5}\,m^2/s$ and $Q_d = 256KJ/mol = 256000\,J/mol$

$$ln\left(\frac{D}{D_0}\right) = -\frac{Q_d}{RT}$$

⇨ $ln(D) - ln(D_0) = -\frac{Q_d}{RT}$

⇨ $T = -\frac{Q_d}{R(ln(D)-ln(D_0))} = -\frac{\frac{256000J}{mol}}{8.31J/mol.k\times(ln(6.5\times10^{-17}m^2/s)-ln(2.7\times10^{-5}m^2/s))}$

⇨ $T = 1151.5\ K - 273 = 878.5°C$

8) The steady-state diffusion flux through a metal plate is $5.4 \times 10^{-10}\ kg/m^2.s$ at a temperature of $727°C$ and when the concentration gradient is $-350\ kg/m^4$. Calculate the diffusion flux at $1027°C$ for the same concentration gradient and assuming an activation energy for diffusion of $125\ KJ/mol$.

Solution:

At a temperature of $727°C(1000K)$:

$$J = -D\frac{\Delta C}{\Delta x}$$

⇨ $D = -\frac{J}{\frac{\Delta C}{\Delta x}} = -\frac{5.4\times10^{-10}kg/m^2.s}{-350kg/m^4} = 1.54 \times 10^{-12}\ m^2/s$

$$D = D_0 \exp\left(-\frac{Q_d}{RT}\right)$$

⇨ $D_0 = D \exp\left(\frac{Q_d}{RT}\right) = (1.54 \times 10^{-12}\ m^2/s) \times \exp\left(\frac{125000\ J/mol}{8.31J/mol.k\times1000K}\right)$

⇨ $D_0 = 5.25 \times 10^{-6}\ m^2/s$

⇨ At a temperature of $1027°C(1300K)$:

⇨ $J = -D\frac{\Delta C}{\Delta x}$

⇨ $J = -D_0 \exp\left(-\frac{Q_d}{RT}\right) \times \frac{\Delta C}{\Delta x}$

⇨ $J = -(5.25 \times 10^{-6}\ m^2/s) \times \exp\left(-\frac{125000\ J/mol}{8.31J/mol.k\times1300K}\right) \times -350\ kg/m^4$

⇨ $J = 1.73 \times 10^{-8}\ kg/m^2.s$

Error function table

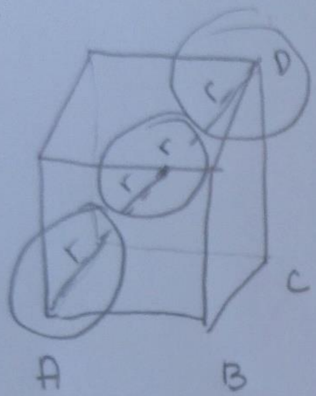
z	erf(z)	z	erf(z)	z	erf(z)
0	0	0.55	0.5633	1.3	0.9340
0.025	0.0282	0.60	0.6039	1.4	0.9523
0.05	0.0564	0.65	0.6420	1.5	0.9661
0.10	0.1125	0.70	0.6778	1.6	0.9763
0.15	0.1680	0.75	0.7112	1.7	0.9838
0.20	0.2227	0.80	0.7421	1.8	0.9891
0.25	0.2763	0.85	0.7707	1.9	0.9928
0.30	0.3286	0.90	0.7970	2.0	0.9953
0.35	0.3794	0.95	0.8209	2.2	0.9981
0.40	0.4284	1.0	0.8427	2.4	0.9993
0.45	0.4755	1.1	0.8802	2.6	0.9998
0.50	0.5205	1.2	0.9103	2.8	0.9999

Diffusion data table

Diffusing Species	*Host Metal*	$D_0(m^2/s)$	*Activation Energy Q_d*		*Calculated Value*	
			kJ/mol	*eV/atom*	*T(°C)*	$D(m^2/s)$
Fe	α-Fe (BCC)	2.8×10^{-4}	251	2.60	500 900	3.0×10^{-21} 1.8×10^{-15}
Fe	γ-Fe (FCC)	5.0×10^{-5}	284	2.94	900 1100	1.1×10^{-17} 7.8×10^{-16}
C	α-Fe	6.2×10^{-7}	80	0.83	500 900	2.4×10^{-12} 1.7×10^{-10}
C	γ-Fe	2.3×10^{-5}	148	1.53	900 1100	5.9×10^{-12} 5.3×10^{-11}
Cu	Cu	7.8×10^{-5}	211	2.19	500	4.2×10^{-19}
Zn	Cu	2.4×10^{-5}	189	1.96	500	4.0×10^{-18}
Al	Al	2.3×10^{-4}	144	1.49	500	4.2×10^{-14}
Cu	Al	6.5×10^{-5}	136	1.41	500	4.1×10^{-14}
Mg	Al	1.2×10^{-4}	131	1.35	500	1.9×10^{-13}
Cu	Ni	2.7×10^{-5}	256	2.65	500	1.3×10^{-22}

*Atomic packing factors for BCC and FCC

BCC



Atomic Packing Factor = APC = $\dfrac{V_{atoms}}{V_{cube = unit\ cell}}$

V_{atoms} = # of atoms * V of 1 atom

# of atoms = 8 corners $\times \frac{1}{8}$ atoms/corner + 1 atom

= 2 atoms in BCC unit

V of 1 atoms = $\frac{4}{3}\pi r^3 \Rightarrow V_{atoms} = 2 \times \frac{4}{3}\pi r^3 = \frac{8}{3}\pi r^3$

$V_{unit\ cell} = V_{cube} = a^3$

$AD^2 = AC^2 + CD^2$

$AD = 4r \Rightarrow AD^2 = 16r^2$

$AC^2 = AB^2 + AC^2 = a^2 + a^2 = 2a^2$

$CD^2 = a^2$

$16r^2 = 2a^2 + a^2$

$16r^2 = 3a^2$

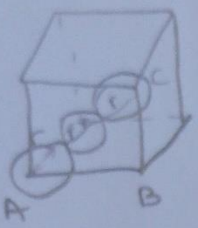
$a^2 = \frac{16}{3} r^2$

$\Rightarrow \boxed{a = \frac{4}{\sqrt{3}} r}$

$APC = \dfrac{\frac{8}{3}\pi r^3}{a^3} = \dfrac{\frac{8}{3}\pi r^3}{\frac{4^3}{3\sqrt{3}} r^3} = \boxed{0.68}$

FCC:

Atomic packing factor $= APF = \dfrac{V_{atoms}}{V_{unit\ cell}}$



$V_{atoms} = \#$ of atoms $*$ V_{atom}

$\#$ of atoms $= \left(\frac{1}{2}\ \text{atoms/face} * 6\ \text{faces}\right) + \left(\frac{1}{8}\ \text{atom/corner} * 8\ \text{corners}\right)$

$= 4$ atoms

$V_{1atom} = \frac{4}{3}\pi r^3 \Rightarrow V_{atoms} = 4 \times \frac{4}{3}\pi r^3 = \frac{16\pi r^3}{3}$

$V_{unit\ cell} = V_{cube} = a^3.$

$AC^2 = AB^2 + BC^2 \Leftrightarrow (4r)^2 = a^2 + a^2 \Rightarrow 16r^2 = 2a^2.$

$a^2 = 8r^2 \Rightarrow a = \sqrt{8}\, r = 2\sqrt{2}\, r.$

$\Rightarrow V_{unit\ cell} = 16\sqrt{2}\, r^3$

$\Rightarrow APF_{FCC} = \dfrac{\frac{16\pi r^3}{3}}{16\sqrt{2}\, r^3} = 0.74$